

MIGA E&S; Portal 2.0

Complete Technical Documentation

MIGAPortal.ESReviewDocumentation.API

All Endpoints · Flow Diagrams · Business Logic · Architecture · Configuration

Project	MIGAPortal.ESReviewDocumentation.API — MIGA E&S; Portal Backend
Framework	.NET 8 / ASP.NET Core Web API
Architecture	CQRS + MediatR + Clean Architecture + Autofac DI
Database	PostgreSQL (Npgsql + EF Core 7)
Auth	Azure AD Bearer JWT + PolicyAuthorize
Dev Base URL	https://localhost:44379/api/esrd
Prod Base URL	https://migaportalclouddev.worldbank.org/api/esrd
Controllers	21 controllers, 130+ endpoints
Services	80+ business logic services
External APIs	14 external integrations
Document	Complete Technical Documentation
Date	2026

1. System Overview

The **MIGA E&S; Portal 2.0** is an enterprise-grade Environmental & Social Risk Management system for the **Multilateral Investment Guarantee Agency (MIGA)**, World Bank Group. It manages the complete lifecycle of MIGA-guaranteed investment projects — from initial pipeline due diligence through active portfolio monitoring, compliance tracking, and reporting.

Application	MIGAPortal.ESReviewDocumentation.API — ASP.NET Core 8 REST API
Purpose	Environmental & Social Due Diligence, Monitoring, ESAP tracking, GBV assessment, Reporting
Users	E&S; Specialists, Managers, GBV Specialists, Administrators — World Bank Group staff
Projects Managed	243 Portfolio Projects, 130+ Pipeline Projects, 181 Shelved Projects
Endpoints	130+ REST API endpoints across 21 controllers
Reports	18 downloadable Excel/PDF reports via Aspose.Cells + Aspose.PDF
Integrations	14 external services: MIGA Portal, Azure AD, Service Bus, Malena AI, Power BI, WbDocs

1.1 Technology Stack

Layer	Technology	Version	Purpose
API Framework	ASP.NET Core	8.0	RESTful API host — controllers, middleware, routing, serialization
Messaging Bus	MediatR	12.x	CQRS pattern — decoupled Query/Command dispatch without direct service calls
DI Container	Autofac	7.1.0	Advanced dependency injection — overrides default ASP.NET Core DI
ORM / Database	EF Core + Npgsql	7.x	PostgreSQL database access via Entity Framework Core
Object Mapping	AutoMapper	12.x	30 mapping profiles — eliminates manual property copying
Validation	FluentValidation	11.3.0	25 validators — run automatically before handlers execute
Report Engine	Aspose.Cells + PDF	-	Excel (.xlsx) and PDF report generation for all 18+ reports
Authentication	Azure AD	JWT Bearer	World Bank SSO — validates tokens against Azure AD
Authorization	PolicyAuthorize	-	ESRD permission codes: ReadAll, EditAll, DashboardManager
AI Analysis	Malena AI (IFC)	-	Sentence-level E&S document analysis — malenaapiext.ifc.org
Event Bus	Azure Service Bus	-	Domain events between MIGA microservices
File Storage	Azure Blob Storage	-	migaportaldevstrg.blob.core.windows.net
Documents	World Bank WbDocs	-	wbdocsapi.ocappsdev.worldbank.org — confidential document store
Analytics	Power BI Embedded	-	ES Insights analytics dashboard

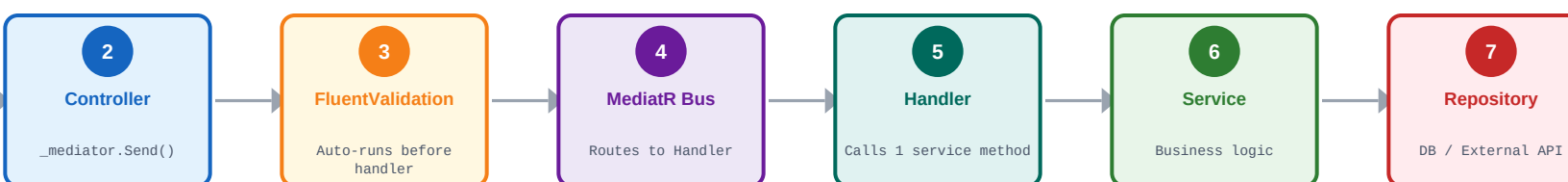
Layer	Technology	Version	Purpose
Secrets	Azure Key Vault	-	migaportal-dev-kvd — DB strings, API keys, never in code
Monitoring	App Insights	-	TelemetryClient — logging, traces, events, exceptions
Resilience	Polly	-	3-retry policy on Malena AI HTTP calls

2. Architecture — CQRS + MediatR Pattern

Every single API operation follows the same CQRS (Command Query Responsibility Segregation) pattern using MediatR as the internal message bus. This is the most important concept to understand before reading the endpoint documentation.

2.1 The Three Layers of Every Request

CQRS Request Flow — How Every Request Travels Through the System



2.2 CQRS: Queries vs Commands

Type	When Used	Changes DB?	Examples
Query	Fetching / reading data	NO	GetPortfolioProjects, GetMonitoringDetails, GetDueDiligenceDetails
Command	Creating / updating / deleting	YES	SaveMonitoringActivity, DeleteEsapAction, SaveDueDiligence

2.3 Standard API Response Envelope

Every endpoint returns the same JSON envelope. Clients should ALWAYS check STATUS_CODE first, not just HTTP status:

```

// JSON – Standard Success Response
{
  "STATUS": "Success",           // Always "Success" by convention
  "STATUS_CODE": 0,             // 0 = success | 500 = error
  "NOTIFICATIONS": [],          // Error/info messages if any
  "Result": [ ... ]             // The actual data payload
}

// JSON – Error Response (e.g. EsId not found)
{
  "STATUS": "Success",           // Still "Success" by convention
  "STATUS_CODE": 500,           // 500 = error occurred
  "NOTIFICATIONS": [
    { "Code": "500", "Message": "EsId not found", "NotificationType": 500 }
  ]
}
  
```

```
],
  "Result": null
}
```

■ **INFO:** File download endpoints (*DownloadXxxReport*) do NOT return JSON. They return *FileStreamResult*: a binary .xlsx stream with *Content-Disposition* header containing the filename.

2.4 Permission Codes Used Across All Endpoints

Permission Code	Who Has It	Endpoints Protected
ReadAll	All authenticated ESRD users	All GET endpoints, all read-only POST/GET queries
EditAll	Specialists with edit rights + Managers/Admins	All create/update/delete/upload endpoints
DashboardManager	Managers and Admins only	Dashboard endpoints showing ALL specialists' data
DashboardSpecialistView	Individual Specialists	Dashboard endpoints scoped to own projects only
InsightsDashboardView	ES Insights users	Power BI and sentence-wise Malena AI endpoints
None / commented out	Any authenticated user	Snapshot endpoints, BgJob endpoints (called by scheduler)

3. Complete Request Flows — End to End

This section documents the complete data flow for the most important operations in the system. Each flow shows every layer from the browser clicking a button to the database and back to the rendered response.

3.1 Portfolio Projects — Complete Sequence Flow

This is the most complex and critical flow in the system. When a user navigates to the Portfolio Projects tab, this exact sequence of operations executes:

1

UI

User Clicks "Portfolio Projects"

React.js triggers HTTP POST /api/esrd/Projects/GetPortfolioProjects with Authorization: Bearer header + JSON body {FiscalYear:[], RegionKey:[], ManagementGroupCode:[], ESMFiscalYear:[],

2

MIDDLEWARE

Azure AD JWT Token Validation

UseAuthentication() middleware validates Bearer token. Checks signature, expiry, audience. Invalid = HTTP 401 Unauthorized returned immediately.

3

CONTROLLER

ProjectsController.GetPortfolioProjects()

[PolicyAuthorize(ESRD, ReadAll)] checked. Body deserialized to PortfolioProjectsQuery. Controller calls _mediator.Send/projectsQuery) — ZERO business logic here.

4

MEDIATR

MediatR Routes to PortfolioProjectsQueryHandler

MediatR lookup: IRequestHandler<PortfolioProjectsQuery, PortfolioProjectsResponse> → PortfolioProjectsQueryHandler. Resolved via Autofac DI container.

5

HANDLER

PortfolioProjectsQueryHandler.Handle()

Handler calls IPortfolioProjectsService.GetPortfolioProjects(request). Awaits result. Wraps in PortfolioProjectsResponse{Result=list, STATUS_CODE=0}.

6

SERVICE

Step 1: Get Portfolio Phase Code from Cache

_commonService.GetEsMasterData(ProjectPhase): fetch the integer code for "Portfolio" phase from in-memory cache (30-min TTL). Used to filter DB query.

7

SERVICE

Step 2: Build LINQ Filter Predicate

BuildEsTablePredicateForPortfolioProjects(): dynamically builds expression tree. Applies only filters present in query (FiscalYear, RegionKey, ManagementGroupCode, etc.).

8

REPOSITORY

Step 3: Query PostgreSQL — Get Portfolio Projection

_projectsRepository.GetPortfolioInformation(predicate, true): EF Core executes SELECT from es_header with JOINS on monitoring, ESRR ratings, team info.

9

EXTERNAL

Step 4: Call MIGA Portal Integration Service

_pipelineMigaPortalIntegrationDataRepository.GetPortfolioProjects(): HTTP GET to MPMIntegrationServiceUrl. Gets all project numbers from MIGA Portal. Finds unassigned projects

10

SERVICE

Step 5: Parallel Data Fetch (Task.WhenAll)

Simultaneously: (a) GetProjectAndApplicationDetailsFromIntegrationService() — full contract/application data, (b) GetProjectsCostCodesFromIntegrationService() — cost codes. Both

11

SERVICE

Step 6: Fetch Team & Lookup Data

_commonService.GetTeamInformation() — team user details. GetEsMasterData(IFCTrack) — IFC Track codes. GetEsMasterData(ESScore) — ES Score lookup tables.

■ **INFO:** The PortfolioProjectsService makes 4 separate external/DB calls and one parallel Task.WhenAll for efficiency. Total typical latency: 800ms–2s depending on project count and Integration Service response time.

13

RESPONSE

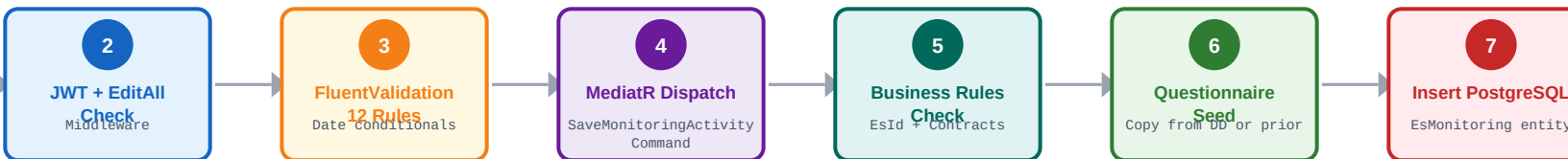
PortfolioProjectsResponse returned to UI

Handler wraps List<PortfolioProjectInformation> in response envelope. ASP.NET Core serializes to JSON (PascalCase). HTTP 200 returned. React renders portfolio grid.

3.2 Save Monitoring Activity — Complete Flow

When a specialist saves a new monitoring activity (AMR review, site visit, or virtual monitoring), this flow executes. It is more complex than a simple save because it includes business rule validation and questionnaire seeding logic:

SaveMonitoringActivity — Multi-Step Save Flow

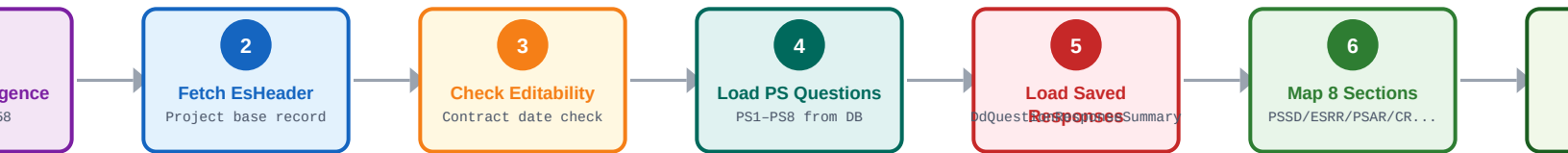


- 1** **UI** **Specialist Fills Monitoring Form**
 User checks AMR Review flag, enters AmrFromDate, AmrToDate, AmrDueDate, AmrReceivedDate. Optionally adds SiteVisit, VirtualMonitoring, ClientMeeting fields. Clicks Save.
- 2** **MIDDLEWARE** **Authentication + Authorization**
 Bearer token validated. [PolicyAuthorize(ESRD, EditAll)] checked. [FromForm] multipart body deserialized to SaveMonitoringActivityCommand.
- 3** **VALIDATOR** **FluentValidation — 12 Conditional Rules**
 If AmrReviewFlag=true: AmrFromDate, AmrToDate, AmrDueDate, AmrReceivedDate ALL required. If SiteVisitFlag=true: SiteVisitFromDate+ToDate required. If ClientMeetingFlag=true:
- 4** **SERVICE** **Verify EsId Exists in Database**
 GetAsync<EsHeader>(x => x.Id == esId). If record not found: add notification "EsId not found", return 0 (error). Handler returns STATUS_CODE=500.
- 5** **SERVICE** **Verify Active Contracts Exist**
 _commonService.CheckIfProjectContainsContracts(projectNumber). HTTP call to Integration Service. No active contracts = monitoring cannot be saved. Return 0.
- 6** **SERVICE** **Map Command to EsMonitoring Entity**
 AutoMapper: MonitoringMapperProfile maps SaveMonitoringActivityCommand → EsMonitoring. ProjectStageId is IGNORED (auto-set by DB on INSERT).
- 7** **SERVICE** **Questionnaire Seeding (First-Ever Record)**
 If NO prior monitoring exists: copy DD questionnaire responses to Monitoring (first monitoring inherits DD assessment). If prior records exist: copy most recent monitoring responses (by
- 8** **REPOSITORY** **Insert EsMonitoring to PostgreSQL**
 EF Core InsertAsync<EsMonitoring>(). Auto-sets CreatedBy (from JWT user UPI), CreatedOn timestamp, IsDeleted=false. Returns new ProjectStageId.
- 9** **RESPONSE** **Return SaveMonitoringActivityResponse**
 Handler wraps: {ProjectStageId: 4521, STATUS_CODE: 0, STATUS: "Success"}. React refreshes monitoring list. Dashboard AMR counters auto-update.

3.3 Get Due Diligence Details — Questionnaire Assembly Flow

GetDueDiligenceDetails is one of the most complex read operations. It assembles a complete multi-section questionnaire from multiple database tables:

GetDueDiligenceDetails — 12-Step Questionnaire Assembly

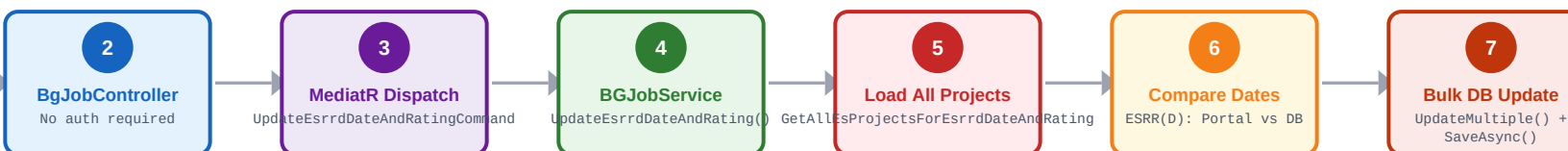


Step	Section Assembled	Data Source	Returned Object
1	EsHeader base info	PostgreSQL: es_header table	EsId, EsStage, ProjectStageId, IsDueDiligenceEditable
2	PS1–PS8 Applicability	DB: ps_question_master + question_options	PsQuestionnaires[] — applicability flags + sub-questions
3	ESRR Factor Scoring	DB: saved responses	EsrrQuestionnaires[] — 10 factor questions with scores 1/2/3
4	PS Achievement Rating	DB: PSAR question master	PsarQuestionnaires[] — PS achievement ratings
5	ESRR Factor Rating Matrix	DB: esrr_factor_rating	EsrrFactorRating — weight matrix for score calculation
6	Contextual Risk Questions	DB: 3-level hierarchy, ordered by SequenceNumber	ContextualRiskQuestionnaires[] — country/sector risk factors
7	E&S Input Section	DB: ps_question_master (Section=EsInput)	EsInput — narrative commentary fields
8	Bubble Information	DB: bubble_info table	BubbleInformation[] — help text tooltips per field

3.4 Background Job Flow — Automated ESRR Date Sync

Background jobs run on a schedule triggered by the World Bank Background Job Scheduler calling BgJobController endpoints. The UpdateEsrrdDateAndRating job is the most important:

Background Job — UpdateEsrrdDateAndRating Flow



Job	Trigger Endpoint	Frequency	What It Does
UpdateEsrrdDateAndRating	POST /BgJob/UpdateEsrrdDateAndRating	Daily	Syncs ESRR(D) dates and ratings from MIGA Portal
UpdateEsrrmDateAndRating	POST /BgJob/UpdateEsrrmDateAndRating	Daily	Syncs ESRR(M) dates and ratings for all portfolio pro
UpdateAmrDueDate	POST /BgJob/UpdateAmrDueDate	Weekly	Recalculates AMR due dates based on contract data
UpdateAmrDueDateAndKnowledgeGap	POST /BgJob/UpdateAmrDueDateAndKnowledgeGap	Weekly	Combined: AMR due date + knowledge gap flag reca
UpdateKnowledgeGap	POST /BgJob/UpdateKnowledgeGap	Weekly	Flags projects approaching or past knowledge gap th
UpdateProjectPhase	POST /BgJob/UpdateProjectPhase	On event	Triggered by ProjectPhaseUpdateEvent from MIGA P
UpdateEsapRequiredData	POST /BgJob/UpdateEsapRequiredData	Weekly	Auto-sets ESAP required flag based on ESRR score
UpdateEsrrDateAndRatingToEsHeader	POST /BgJob/UpdateEsrrDateAndRatingToEsHeader	Daily	Syncs ESRR dates to EsHeader table for fast lookup
CreateAmrHistory	POST /BgJob/CreateAmrHistory	Monthly	Creates AMR history records for all portfolio projects
UpdateAmrCalendarYear	POST /BgJob/UpdateAmrCalendarYear	Annually	Migrates AMR records from fiscal year to calendar ye
GetMigaWeeklySnapshot	GET /BgJob/GetMigaWeeklySnapshot	Weekly	Generates weekly summary Excel for each specialist

4. Project Lifecycle — From Application to Archive

Every MIGA-guaranteed project passes through a defined lifecycle managed by the E&S; Portal. Understanding this lifecycle is essential for understanding why each endpoint exists and when it is called.

1. Application Received	Project enters the system as a pipeline project. Metadata recorded: Project ID, Registration Number, Region, Host Country, Investor Country, Guarantee Holder, Sector, Sub-Sector, Management Group, Product Type, Fiscal Year. Initial status: PIPELINE ACTIVE, LEAD, or SLOW MOVING.
2. E&S; Specialist Assignment	Managers assign an E&S; Specialist (and optionally Support Specialist, Manager, Peer Reviewer) to the project via SaveTeamInformation endpoint. Unassigned projects appear in the 'Unassigned Projects' sub-tab with 16+ projects awaiting assignment.
3. Environmental & Social Due Diligence	Specialist conducts full ESDD (Environmental & Social Due Diligence) assessment. PS1–PS8 applicability determined. Management Factors (3 questions), Performance Factors (5), Communication Factors (2) scored 1–3. Contextual Risk questionnaire completed. ESRR(D) calculated.
4. ESRR(D) — Risk Rating Generation	System calculates Environmental & Social Risk Rating at Due Diligence stage from factor scores. Specialist may override with justification. Manager approves override. Rating codes: Excellent, Satisfactory, Partly Unsatisfactory, Unsatisfactory, NA.
5. ESAP Creation (if required)	If E&S; compliance actions are required, specialist creates ESAP (Environmental & Social Action Plan) items with due dates, deliverables, responsible parties, and remarks. Guarantee holder is contractually bound to complete all ESAP actions.
6. GBV Risk Assessment	E&S; Specialist completes the Gender-Based Violence risk questionnaire. System calculates GBV Risk Score and Risk Rating (Low FI, Medium FI, High FI, etc.). Rating status: Final, Pending, or Not Required.
7. Document Upload	Supporting documents uploaded: ESIA's (Environmental & Social Impact Assessments), audit logs, ESAP templates, risk input forms. Documents stored in World Bank WbDocs system with Strictly Confidential classification.
8. Management Review & Guarantee Issuance	MIGA management reviews E&S; assessment. Approved projects receive MIGA guarantee and move to Portfolio. Declined or withdrawn projects move to Shelved. Portfolio transition triggers ProjectPhaseUpdateEvent on Azure Service Bus.
9. Ongoing Portfolio Monitoring	Portfolio project undergoes regular E&S; monitoring throughout guarantee life. Monitoring activities: AMR Review (Annual Monitoring Report), Site Visit (physical or virtual), Client Meeting, Other. ESRR(M) updated each monitoring cycle.

10. ESAP Completion Tracking

Specialist verifies guarantee holder completion of ESAP actions. Completion dates recorded, deliverables verified. Missed deadlines trigger automated email reminders via EsapNotificationService.

11. High Risk Classification

If major E&S issues arise, specialist flags project as High Risk. CAO (Compliance Advisor Ombudsman) complaint status tracked. Rationale, key issues, and recent developments documented. Triggers inclusion in High Risk reports.

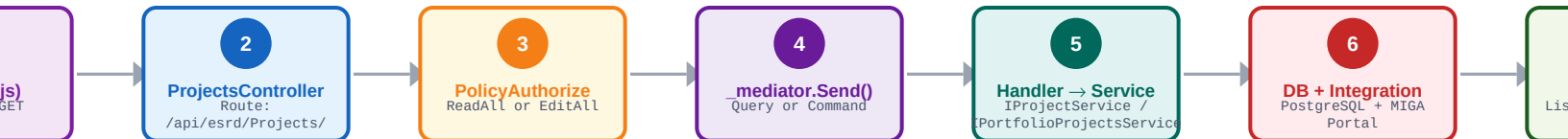
12. Project Closure / Archive

When guarantee expires, project is terminated or withdrawn → moves to Shelved Projects (181 records). All data retained read-only for compliance. Status codes: DECLINED, WITHDRAWN, NOT GOING FORWARD, DESELECTED BY MIGA.

5. Projects API — /api/esrd/Projects

The Projects controller is the largest in the system with 24+ endpoints covering the complete project lifecycle: fetching pipeline, portfolio, and shelved projects; team management; GBV assessment; AMR history; and all project-level file downloads.

Projects Controller — Request Flow Pattern



5.1 Read Endpoints (Queries)

POST GetPipelineProjects

[ReadAll](#)

Fetch all projects in pipeline stage. Filter body: FiscalYear[], ManagementGroupCode[], ProductType[], RegionKey[], HostCountryKey[], ProjectCategory[], BusinessStage[], EsSpecialist[]. All filters are optional — omit for all projects. Returns List<PipelineProjectInformation> with DD stage, team assignment, ESRR(D) score, GBV rating, and all project metadata.

REQUEST PARAMETERS

<code>FiscalYear</code>	<code>int[]</code>	optional — e.g. [2025, 2024]
<code>RegionKey</code>	<code>string[]</code>	optional — e.g. ['AFR','EAP']
<code>EsSpecialist</code>	<code>int[]</code>	optional — UPI list for filtering

RETURNS: PipelineProjectsResponse { Result: List<PipelineProjectInformation> }

POST GetPortfolioProjects

[ReadAll](#)

Fetch all actively monitored portfolio projects. Same filter structure as pipeline plus ESMFiscalYear[]. Internally merges data from ESRD PostgreSQL DB with MIGA Portal Integration Service. Returns 80+ fields per project including AMR dates, ESRR(M) scores, MSV data, team details, and contract information.

REQUEST PARAMETERS

<code>FiscalYear</code>	<code>int[]</code>	optional
<code>ESMFiscalYear</code>	<code>int?[]</code>	optional — ESM fiscal year filter
<code>ManagementGroupCode</code>	<code>int[]</code>	optional

RETURNS: PortfolioProjectsResponse { Result: List<PortfolioProjectInformation> (80+ fields) }

GET

GetShelvedProjects[ReadAll](#)

Fetch all terminated/shelved projects (181 records). Read-only — no edit operations permitted. Returns status codes: DECLINED, WITHDRAWN, NOT GOING FORWARD, DESELECTED BY MIGA.

REQUEST PARAMETERS

`search params` `query string` `EsId`, `ProjectNumber`, `Status`

RETURNS: `ShelvedProjectsResponse { Result: List<ShelvedProjectInformation> }`

GET

GetProjectDetails[ReadAll](#)

Fetch complete details for a single project. Works for pipeline, portfolio, and shelved. Returns all tab data: project info, team assignment, contract details, GBV score, ESRR scores.

REQUEST PARAMETERS

`EsId` `int` required — internal E&S ID
`IsProjectsInPipeline` `bool` true for pipeline
`IsShelvedProject` `bool` true for shelved

RETURNS: `ProjectDetailsResponse { full project object }`

POST

GetGbvDetails[ReadAll](#)

Fetch full GBV (Gender-Based Violence) risk assessment questionnaire for a project. Returns structured questionnaire with risk-rated responses, GBV Risk Score, and GBV Risk Rating.

REQUEST PARAMETERS

`EsId` `int` required
`GbvId` `int` 0 if new assessment
`HostCountryKey` `string` for country-level risk data
`IsProjectInPipeline` `bool` scopes the assessment stage

RETURNS: `GbvDetailsResponse { questionnaire + risk score + rating }`

5.2 Write Endpoints (Commands)

POST

SaveTeamInformation[EditAll](#)

Assign E&S team to a project. Creates or updates the team assignment record. Sends team assignment notification email. Triggers domain event for cross-portal sync.

REQUEST PARAMETERS

<code>EsId int</code>	required — project to assign team to
<code>EsSpecialistUpI int</code>	required — lead specialist UPI
<code>SupportSpecialistUpIList optional[]</code>	optional — co-specialists
<code>ProjectManager int</code>	optional — manager UPI

RETURNS: `SaveTeamInformationResponse { STATUS_CODE: 0, team confirmation }`

POST

SaveGbvQuestionnaires[EditAll](#)

Save GBV questionnaire answers and calculate GBV Risk Rating. Stores all question responses and persists the calculated GBV score and risk rating for the project.

REQUEST PARAMETERS

<code>EsId int</code>	required
<code>GbvId int</code>	0=new, >0=update
<code>GbvRiskRatingStatusCode int</code>	Final/Pending/NotRequired
<code>Questionnaires array</code>	question-value pairs

RETURNS: `SaveGbvQuestionnairesCommandResponse { GbvId, STATUS_CODE: 0 }`

POST

SaveGbvActionPlanDetail[EditAll](#)

Create or update a GBV action plan item. Links mitigation actions to the project's GBV risk assessment.

REQUEST PARAMETERS

<code>GbvActionPlanDetailId int</code>	0=new
<code>GbvId int</code>	parent GBV assessment ID
<code>TypeOfAction string</code>	action category
<code>ExpectedOutput string</code>	deliverable
<code>DueDate datetime</code>	target completion date

RETURNS: `SaveGbvActionPlanDetailResponse { actionPlanDetailId }`

POST

SaveAdditionalInformationForPortfolio

EditAll

Save additional metadata fields specific to portfolio projects: Track B classification, knowledge gap flags, ESM indicators, custom annotations not in the base project record.

REQUEST PARAMETERS

EsId int required
 IsIfcTrackB bool IFC Track B flag
 IsKnowledgeGapActionPlan KGflag

RETURNS: SaveResponse { STATUS_CODE: 0 }

5.3 Download Endpoints

All download endpoints follow the same pattern: they receive filter parameters, generate an Excel file via Aspose.Cells, and return a **FileStreamResult** (not JSON). The browser automatically triggers a Save File dialog.

Endpoint	Filter Parameters	Output File
DownloadPipelineProjectsReport	Same as GetPipelineProjects filters	Pipeline_Projects_Report_YYYYMMDD.xlsx
DownloadPortfolioProjectsReport	Same as GetPortfolioProjects filters	Portfolio_Projects_Report_YYYYMMDD.xlsx
DownloadPipelineProjectsSnapshotReport	FiscalYear, date range	Pipeline_Snapshot_YYYYMMDD.xlsx
DownloadPortfolioProjectsSnapshotReport	FiscalYear, date range	Portfolio_Snapshot_YYYYMMDD.xlsx
DownloadShelvedProjectsReport	Status filter	Shelved_Projects_Report_YYYYMMDD.xlsx
DownloadGbvActionPlanDetailsReport	GbvId, EsId	GBV_Action_Plan_EsId_YYYYMMDD.xlsx
DownloadProjectsGbvDetailsReport	FiscalYear, specialist	Projects_GBV_Details_YYYYMMDD.xlsx

6. Monitoring API — /api/esrd/Monitoring

The Monitoring controller manages all ongoing E&S; monitoring activities for portfolio projects. Monitoring is how MIGA tracks whether guarantee holders are maintaining adequate E&S; performance after the guarantee is issued. Activities include Annual Monitoring Reports (AMR), site visits, virtual monitoring, and client meetings.

Monitoring Save Flow — SaveMonitoringActivity

**GET**

GetMonitoringDetails

[ReadAll](#)

Retrieve full monitoring questionnaire detail for a specific monitoring activity. Returns the complete questionnaire with all section responses, ESRR(M) scores, factor comments, and monitoring activity metadata.

REQUEST PARAMETERS

`EsId int` required — project ID
`ProjectStageId int` required — specific monitoring record ID

RETURNS: `MonitoringQuestionnaireResponse { questionnaire with all section data }`

GET

GetMonitoringList

[ReadAll](#)

Retrieve chronological list of all monitoring activities for a project. Each record shows monitoring type flags (AMR/SiteVisit/Virtual/Client), dates, and ESRR(M) rating at the time of monitoring.

REQUEST PARAMETERS

`EsId int` required — project ID

RETURNS: `MonitoringListResponse { Result: List<MonitoringActivityInformation> }`

GET

GetDueDateForProject

ReadAll

Calculate and return the AMR (Annual Monitoring Report) due date for a portfolio project. Based on contract effective date, AMR frequency setting (annual/biannual), and last received AMR date.

REQUEST PARAMETERS

EsId int required — project ID

RETURNS: MonitoringDueDateResponse { DueDate, Frequency, LastReceivedDate }

POST

SaveMonitoringActivity

EditAll

Create or update a monitoring activity. [FromForm] multipart: supports file attachments. Applies 12 conditional validation rules. Automatically seeds monitoring questionnaire from DD responses on first activity. Core operation for all monitoring workflow.

REQUEST PARAMETERS

ProjectStageId int 0=new record, >0=update existing
EsId int required
FiscalYear int MIGA fiscal year
AmrReviewFlag bool? AMR review conducted?
AmrDueDate datetime? required if AmrReviewFlag=true

RETURNS: SaveMonitoringActivityResponse { ProjectStageId: int (new/updated record ID) }

POST

SaveMonitoringActivityAndAmrHistory

EditAll

Combined atomic save: creates monitoring activity AND AMR history record in a single database transaction. Used for the primary monitoring workflow where AMR review must be tracked historically.

REQUEST PARAMETERS

monitoring fields same as SaveMonitoringActivity
amr history AmrFromDate, AmrSpecificDate, AmrReceivedDate, AmrReviewedDate

RETURNS: SaveMonitoringActivityAndAmrHistoryResponse { ProjectStageId, AmrHistoryId }

POST

DeleteMonitoring

EditAll

Soft-delete a monitoring activity record. [FromQuery] parameter. The record is marked as deleted (IsDeleted=true) but not physically removed from the database.

REQUEST PARAMETERS

ProjectStageId int (query required) monitoring record to delete

RETURNS: SaveResponse { STATUS_CODE: 0 }

POST

DownloadMonitoringReport

ReadAll

Generate and download the monitoring questionnaire as a formatted Excel report. Includes all questionnaire sections, ESRR(M) scores, factor comments, and monitoring metadata.

REQUEST PARAMETERS

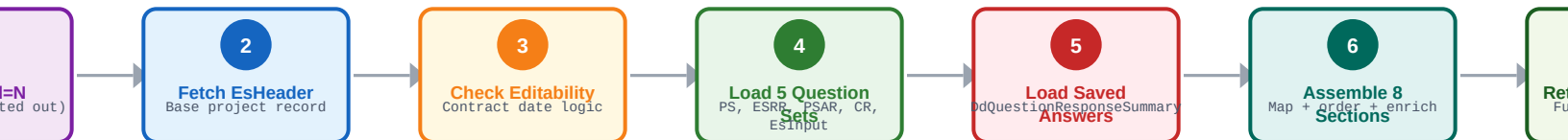
DownloadMonitoringReportQueryId, ProjectStageId, and filter params

RETURNS: FileStreamResult (.xlsx binary, Content-Disposition: attachment)

7. Due Diligence API — /api/esrd/DueDiligence

The Due Diligence controller manages the E&S; Due Diligence questionnaire — the core assessment tool for pipeline projects. The DD questionnaire evaluates a project against IFC Performance Standards PS1–PS8 and generates the ESRR(D) risk rating. It is one of the most data-rich endpoints in the entire system.

Due Diligence — GetDetails Assembly Flow



7.1 DD Questionnaire Structure

Section	Description	Data Source	Questions
PssdQuestionnaires	PS1–PS8 applicability assessment. Is each Performance Standard applicable to this project? Yes/No/NA.	PerformanceStandard + DDQuestionnaireResponse	10
EsrrQuestionnaires	ESRR factor scoring. Each factor rated 1 (Poor), 2 (Average), 3 (Good), 4 (Excellent), 5 (Performance + 2 Co).	DDQuestionnaireResponse	5
PsarQuestionnaires	PS Achievement Rating — how well is the project meeting each applicable PS requirement?	DDQuestionnaireResponse	8
ContextualRiskQuestionnaires	Country and sector-specific risk factors. Three contextual risk factors identified by multiple users/country	DDQuestionnaireResponse	3
EsInput	Free-form narrative comments for E&S Specialist assessment (Section 7.3 field)	DDQuestionnaireResponse	1
ResponseSummary	Summary of all scores: ESRR(D) rating code, DDQuestionnaireResponse, ESRR analysis, raw score, all factor comments	DDQuestionnaireResponse	1
EsrrFactorRating	Weight matrix used to calculate overall ESRR score from rating data	DDQuestionnaireResponse	1
BubbleInformation	Help text tooltips displayed next to each question bubble in the UI.	DDQuestionnaireResponse	1

GET GetDueDiligenceDetails

[None \(open\)](#)

Assemble and return the complete DD questionnaire for a project. Returns all 8 questionnaire sections with saved responses, calculated ESRR(D) rating, editability flag based on contract date, and tooltip bubble information.

REQUEST PARAMETERS

`EsId` `int` (query string) required — project to fetch questionnaire for

RETURNS: `DueDiligenceQuestionnaireResponse` { 8 section arrays + `ResponseSummary` + `IsDueDiligenceEdi`

POST**SaveDueDiligence**[EditAll](#)

Save all DD questionnaire responses in a single call. Validates editability check (cannot edit if active contracts exist). Handles ESRR override — if DeleteEsrrOverridenRatingInfo=true, clears override fields. Saves GeneralInformation summary AND all individual QuestionDetails in one transaction.

REQUEST PARAMETERS

EsId int	required
ProjectStageId int	0=new, >0=update
GeneralInformation object	ESRR scores, dates, override, all factor comments
QuestionDetails array	each: QuestionId, OptionId, TextBoxAnswer, CheckBoxChecked, QuestionSection
IsDueDiligenceEditable bool	from frontend editability flag

RETURNS: SaveDueDiligenceResponse { ProjectStageId: int }

POST**DeleteDueDiligence**[EditAll](#)

Soft-delete a due diligence record. Removes the DD assessment for the specified project stage. The data is marked IsDeleted=true but retained for audit purposes.

REQUEST PARAMETERS

ProjectStageId int	(query required) DD record to delete
---------------------------	--------------------------------------

RETURNS: SaveResponse { STATUS_CODE: 0 }

POST**DownloadDueDiligenceReport**[ReadAll](#)

Generate and download the complete DD questionnaire as a formatted Excel or PDF report. Includes all questionnaire sections, PS assessments, ESRR scores, factor comments, and narrative sections. Used for MIGA management review submissions.

REQUEST PARAMETERS

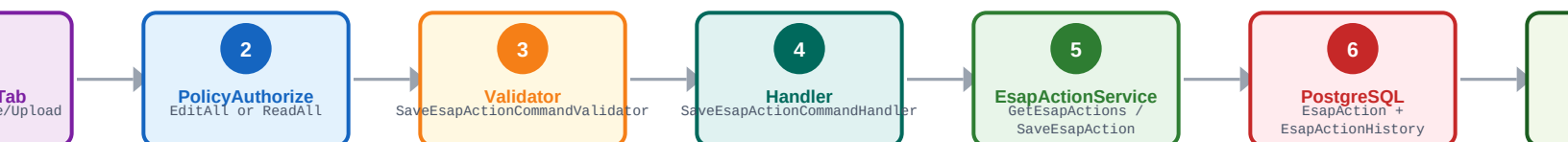
full questionnaire JSON	DownloadDueDiligenceReportQuery with all questionnaire data
--------------------------------	---

RETURNS: FileStreamResult (.xlsx or .pdf binary)

8. ESAP Action API — /api/esrd/EsapAction

The ESAP Action controller manages Environmental & Social Action Plans — contractual lists of actions that MIGA guarantee holders must complete. Actions have due dates, deliverables, completion dates, and a full audit history showing who changed what and when.

ESAP Action API — Complete Flow



GET GetEsapActions

[ReadAll](#)

Retrieve all ESAP actions for a project with full history. For each action, fetches the complete `EsapActionHistory[]` including who made each change (user name resolved from UPI). Also returns `IsEsapRequired` flag and the ESAP question text.

REQUEST PARAMETERS

`EsId int (query string)` required — project ID

RETURNS: `EsapActionsResponse { EsapRequiredInfo{IsEsapRequired, Question}, EsapActions[ {...EsapActi`

GET GetEsapDetails

[ReadAll](#)

Get full ESAP details in one call: `IsEsapRequired` flag + all actions with history. Same as `GetEsapActions` but returns `EsapDetailsResponse` wrapper.

REQUEST PARAMETERS

`EsId int (query string)` required

RETURNS: `EsapDetailsResponse { IsEsapRequired, EsapActions[] }`

GET GetEsapRequired

[ReadAll](#)

Check only whether ESAP is required for a project, without fetching all actions. Lightweight endpoint for initial tab rendering.

REQUEST PARAMETERS

`EsId int (query string)` required

RETURNS: `EsapRequiredResponse { IsEsapRequired: bool, Question: string }`

POST**SaveEsapAction**[EditAll](#)

Create a new ESAP action or update an existing one. Creates an EsapActionHistory entry automatically recording who saved it and when. Sets ProjectStageType from EsHeader.EsStage.

REQUEST PARAMETERS

<code>EsapActionId</code>	<code>int</code>	0=new action, >0=update existing
<code>EsId</code>	<code>int</code>	required
<code>Action</code>	<code>string</code>	action description text
<code>DueDate</code>	<code>datetime</code>	required — target completion date
<code>RescheduleDate</code>	<code>datetime</code>	optional — new date if rescheduled

RETURNS: `SaveEsapActionResponse { EsapActionId: int }`

POST**SaveEsapRequired**[EditAll](#)

Toggle whether ESAP is required for a project. Updates EsHeader.IsEsap field. Simple boolean toggle.

REQUEST PARAMETERS

<code>EsId</code>	<code>int</code>	required
<code>IsEsapRequired</code>	<code>bool</code>	true to require ESAP, false to remove requirement

RETURNS: `SaveEsapRequiredResponse { STATUS_CODE: 0 }`

POST**DeleteEsapAction**[EditAll](#)

Delete one or more ESAP action items. Accepts a list of IDs for bulk delete. Soft-deletes records (IsDeleted=true). Requires confirmation in UI before calling.

REQUEST PARAMETERS

<code>esapActionIdList</code>	<code>int[]</code>	required — list of action IDs to delete e.g. [1,2,3]
-------------------------------	--------------------	--

RETURNS: `SaveResponse { STATUS_CODE: 0, count of deleted records }`

POST**UploadEsapAction**[EditAll](#)

Upload ESAP actions from the Excel template file. [FromForm] multipart. Aspose.Cells reads the uploaded .xlsx, extracts action rows, deletes existing non-DD ESAP actions, creates new ones, and sets IsEsap=true on the EsHeader record.

REQUEST PARAMETERS

EsId `int` (form field) required
UploadFile `binary (.xlsx)` the filled ESAP_Actions_Template.xlsx file

RETURNS: `SaveResponse { STATUS_CODE: 0, count of uploaded actions }`

POST**DownloadEsapActionTemplate**[None](#)

Download the blank ESAP action plan template. No body required. Reads ESAP_Actions_Template.xlsx from the Templates folder on disk and streams it to the client. Users fill this template and upload via UploadEsapAction.

RETURNS: `FileStreamResult (ESAP_Actions_Template.xlsx binary)`

POST**DownloadEsapActionReport**[ReadAll](#)

Generate formatted ESAP status Excel report for a project. Includes all action items with status (overdue/on-track/completed), due dates, completion dates, deliverables, and history.

REQUEST PARAMETERS

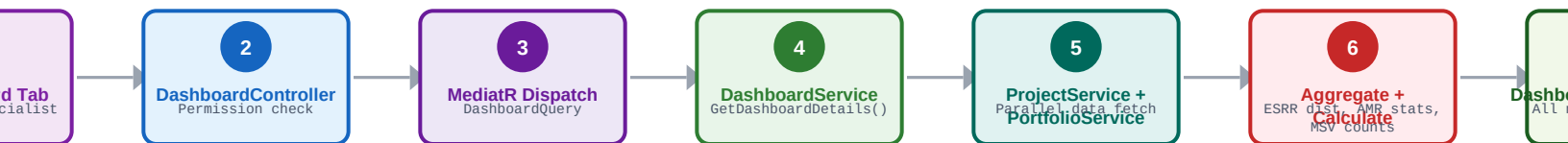
EsId `int` required
action data `JSON body` DownloadEsapActionReportCommand with project and action data

RETURNS: `FileStreamResult (ESAP_Report_ProjectName_YYYYMMDD.xlsx)`

9. Dashboard API — /api/esrd/Dashboard

The Dashboard controller provides aggregated analytics for the E&S; Dashboard landing page. Two permission tiers exist: **DashboardManager** (sees all specialists' data aggregated) and **DashboardSpecialistView** (sees only their own assigned projects). The dashboard combines pipeline, portfolio, monitoring, and reporting data into a single comprehensive view.

Dashboard API — Data Aggregation Flow

**POST**

GetDashboardDetails

DashboardManager

Manager-level dashboard: aggregated data across ALL specialists. Returns complete dashboard analytics: project counts by status, ESRR(D) and ESRR(M) distributions, MSV summary, AMR status, business stage breakdown, sector distribution, management group breakdown, upcoming contract expirations, and unassigned project count.

REQUEST PARAMETERS

`EsSpecialistUPI` `int?` optional — 0 or omit for all specialists
`FiscalYear` `int[]` optional filter
`ManagementGroupCode` `int[]` optional filter

RETURNS: `DashboardDetailsResponse` { pipeline stats, portfolio stats, ESRR distributions, AMR summar

POST

GetDashboardDetailsForSpecialist

DashboardSpecialistView

Specialist-scoped dashboard: same structure as `GetDashboardDetails` but filtered to only show the logged-in specialist's assigned projects. Enables personal workload view.

REQUEST PARAMETERS

same as `GetDashboardDetails` scoped to logged-in specialist's UPI

RETURNS: `DashboardDetailsResponse` { specialist-filtered metrics }

POST

GetPipelineProjectsForDashboardDetails

DashboardManager

Pipeline-specific graphical analytics. Returns chart data for: ESRR(D) distribution donut chart, sector distribution donut, management group bar chart, business stage summary circles, and GBV rating breakdown.

REQUEST PARAMETERS

`FiscalYear` `int[]` optional
`ManagementGroupCode` `int[]` optional
`ProductType` `string[]` optional

RETURNS: `PipelineProjectsDashboardDetailsResponse { distribution data for charts }`

POST

GetPortfolioProjectsForDashboardDetails

DashboardManager

Portfolio-specific graphical analytics. Returns chart data for: ESRR(M) distribution, MSV completion rate donut, AMR submission bar chart, project category distribution, Track B portfolio count.

REQUEST PARAMETERS

`same filters as pipeline` `FiscalYear`, `ManagementGroupCode`, `ProductType`

RETURNS: `PortfolioProjectsDashboardDetailsResponse { portfolio chart data }`

POST

DownloadDashboardReport[ReadAll](#)

Download comprehensive dashboard data as Excel workbook. Contains multiple sheets: Pipeline Summary, Portfolio Summary, Other Initiatives. Used for management reporting.

REQUEST PARAMETERS

`filter params` `JSON body` `FiscalYear`, management group, specialist filters

RETURNS: `FileStreamResult (Dashboard_Report_YYYYMMDD.xlsx)`

10. Reports API — /api/esrd/Reports

The Reports controller provides 19 endpoints for generating and downloading compliance, monitoring, risk, and performance reports. All download endpoints generate Excel (.xlsx) files via Aspose.Cells and return `FileStreamResult`. Report data preview endpoints (Get*) return structured data for the Reports screen UI before the user decides to download.

Reports API — Download Flow Pattern



10.1 Report Data Preview Endpoints (GET — Returns JSON for UI display)

POST GetReportDetails

[ReadAll](#)

Aggregate report data for the Reports screen preview. Returns summary data across all report types: project counts for recently updated, ESAP status summary, ESRR score distributions, AMR submission rates, and MSV trend data. Used to populate the Reports landing page before a user decides to download.

REQUEST PARAMETERS

`filter params` JSON body FiscalYear, ManagementGroupCode, date range filters

RETURNS: `ReportDetailsResponse { projectLists, esapStatus, esrrScores, amrStats }`

GET GetTrendingEsrrReportDetails

[ReadAll](#)

ESRR(M) trend analytics — full detail. Returns how ratings changed quarter by quarter across all portfolio projects. Enables compliance trend analysis and early warning detection.

RETURNS: `TrendingEsrrStatusReportDetailsResponse { quarterly ESRR trend data }`

GET GetTrendingEsrrSummaryReportDetails

[ReadAll](#)

Summarized ESRR trend data used for the trend summary chart on the Reports screen. Lighter payload than full detail endpoint.

RETURNS: `TrendingEsrrStatusSummaryReportDetailsResponse { summary trend chart data }`

GET

GetAmrSubmissionReportDetails

ReadAll

AMR (Annual Monitoring Report) submission analytics. Returns submission rates, outstanding AMRs count, overdue submissions broken down by fiscal year and specialist. Used for compliance performance tracking.

RETURNS: AmrSubmissionReportDetailsResponse { submission rates, outstanding, overdue by specialist }

GET

GetMsvReportDetails

ReadAll

Monitoring Site Visit analytics. Returns MSV completion rates, recommended vs completed visits, Track B portfolio data. Optional FiscalYear filter.

REQUEST PARAMETERS

FiscalYear int (query string optional)

RETURNS: MsvReportDetailsResponse { MSV completion rates, trends }

10.2 Download Report Endpoints (POST — Returns .xlsx FileStreamResult)

Endpoint	Description	Key Output Columns
DownloadHighRiskReport	Full High Risk project detail report	Project Name, Region, CAO Complaint, Rationale, Key Issues, Rece
DownloadHighRiskListReport	Simplified list of all current High Risk projects	Project ID, Name, Region, Specialist, High Risk Date, Guarantee Hol
DownloadHighRiskRemovalReport	Detail report for projects removed from High Risk	All High Risk fields + Removal Date, Removal Rationale
DownloadHighRiskRemovalListReport	Simplified list of removed High Risk projects	Project ID, Name, Removal Date, Previous Rating
DownloadMajorIncidentReport	All major E&S incidents across MIGA projects	Incident Date, Type, Severity, Description, Corrective Actions, Status
DownloadEsapStatusReport	ESAP action compliance status across portfolio	Action Description, Due Date, Status (Overdue/On Track/Completed)
DownloadEsrrStatusReport	ESRR(M) ratings for all portfolio projects	Project ID, Name, ESRR(M) Rating, Score, Date, Previous Rating, C
DownloadAmrSubmissionReport	AMR submission tracking by fiscal year	Project ID, AMR Due Date, Received Date, Outstanding Flag, Overd
DownloadMonitoringVisitTrendsReport	MSV completion trends over fiscal years	FY, Recommended Count, Completed Count, Completion Rate, Trac
DownloadEsrrScoreForPortfolioReport	ESRR(M) scores for all portfolio projects	Project ID, Name, ESRR(D), ESRR(M), Change, Date, Specialist, Ov
DownloadEsrrScoreForIndividualProjectReport	ESRR score history for one project	Date, ESRR Score, Rating, Override Flag, Justification — one row pe
DownloadMonitoringProjectReport	Comprehensive monitoring activities report	All AMR, site visit, virtual monitoring records per project
DownloadNatureFinanceReport	Nature Finance assessments for all projects	Project ID, Potential Score, IsNatureFinance (Y/N), Justification
DownloadRecentlyUpdatedProjectsReport	Projects with changes in last N days	Project ID, Name, Last Modified, Changed By, Change Type

11. High Risk, Nature Finance & Incident APIs

These three controllers manage specialist E&S; risk assessments beyond the standard DD questionnaire.

11.1 HighRisk API — /api/esrd/HighRisk

High Risk classification is applied to portfolio projects with significant E&S; performance issues. It triggers enhanced monitoring requirements, manager review, and inclusion in the High Risk reports. CAO (Compliance Advisor Ombudsman) complaint status is tracked alongside the classification.

GET

GetHighRiskDetails

[ReadAll](#)

Fetch complete High Risk questionnaire for a project. Returns IsHighRisk flag, CAO complaint status, rationale, key E&S issues, and recent developments. If the project is currently NOT high risk, also returns the most recent historical high-risk responses (when it was previously classified as high risk).

REQUEST PARAMETERS

`EsId int (query string)` required

RETURNS: `HighRiskQuestionnaireResponse { IsHighRisk, IsCaoComplaint, Rationale, KeyIssues, RecentDe`

POST

SaveHighRiskDetails

[EditAll](#)

Save or update High Risk classification for a project. The HighRiskService builds a complete questionnaire from the existing DB record, merges in new responses from the command, and persists. Previous high risk summaries are preserved for historical comparison.

REQUEST PARAMETERS

`EsId int` required
`IsHighRisk bool` true to flag as High Risk
`IsCaoComplaint bool` CAO complaint filed?
`Rationale string` reason for classification
`KeyIssues string` specific E&S issues identified

RETURNS: `SaveResponse { STATUS_CODE: 0 }`

11.2 NatureFinance API — /api/esrd/NatureFinance

Nature Finance assessments align MIGA's portfolio with global biodiversity frameworks (TNFD, NGFS). The NatureFinanceService uses an **upsert pattern**: if a record exists for this EsId + EsStage, update it; otherwise insert a new record. Values are stored as UPPERCASE codes internally.

GET

GetNatureFinanceDetails[ReadAll](#)

Fetch Nature Finance assessment for a project. Returns IsNatureFinance code + display text, Potential code + display text, and Justification. If no record exists, returns empty object with blank strings (not an error).

REQUEST PARAMETERS

`EsId` `int` (query string) required
`EsStage` `string` (query string) required — e.g. 'DD' or 'MN'

RETURNS: `NatureFinanceDetailsResponse { IsNatureFinance, IsNatureFinanceText, Potential, PotentialT`

POST

SaveNatureFinance[EditAll](#)

Upsert Nature Finance assessment. Stores IsNatureFinance and Potential as UPPERCASE. Maps code to display text via Constants.NatureFinanceValues and Constants.NatureFinancePotentialValues dictionaries for the response.

REQUEST PARAMETERS

`EsId` `int` required
`EsStage` `string` required — 'DD' or 'MN'
`IsNatureFinance` `string` code: 'Y'/'N'/'NA'
`Potential` `string` code: 'H'/'M'/'L'/'NA'
`Justification` `string` explanation text

RETURNS: `SaveNatureFinanceResponse { STATUS_CODE: 0, updated fields with display texts }`

11.3 Incident API — /api/esrd/Incident

E&S; Incidents are significant environmental or social events at project sites: environmental accidents, community complaints, labor violations, or safety incidents. The IncidentService integrates with WbDocs for document storage.

GET

GetIncidents[ReadAll](#)

List all E&S incidents for a project chronologically.

REQUEST PARAMETERS

`EsId` `int` (query string) required

RETURNS: `IncidentListResponse { Result: List<IncidentSummary> }`

GET

GetIncidentDetail[ReadAll](#)

Full incident detail including incident type, date, description, severity, corrective action plan, and attached documents.

REQUEST PARAMETERS

`IncidentId` `int` (query string) required

RETURNS: `IncidentDetailResponse { full incident object + documents list }`

POST

SaveIncident[EditAll](#)

[FromForm] Create or update an incident. Includes optional COVID-19 detail sub-records and optional document upload directly to WbDocs. Saves incident record first, then COVID details, then processes document uploads sequentially.

REQUEST PARAMETERS

<code>EsId</code> <code>int</code>	required
<code>IncidentDate</code> <code>datetime</code>	required
<code>Description</code> <code>string</code>	required
<code>IncidentType</code> <code>string</code>	classification
<code>Severity</code> <code>string</code>	Low/Medium/High

RETURNS: `SaveIncidentResponse { IncidentId, STATUS_CODE: 0, documentUploadResults }`

POST

DeleteIncident[EditAll](#)

Soft-delete an incident record. Also calls Document DeleteIncident to clean up linked document references.

REQUEST PARAMETERS

`IncidentId` `int` (query string) required

RETURNS: `SaveResponse { STATUS_CODE: 0 }`

POST

UploadDocumentForIncident

EditAll

[FromForm] Upload additional supporting document for an existing incident. Saves to WbDocs document system and creates document metadata record linked to the incident.

REQUEST PARAMETERS

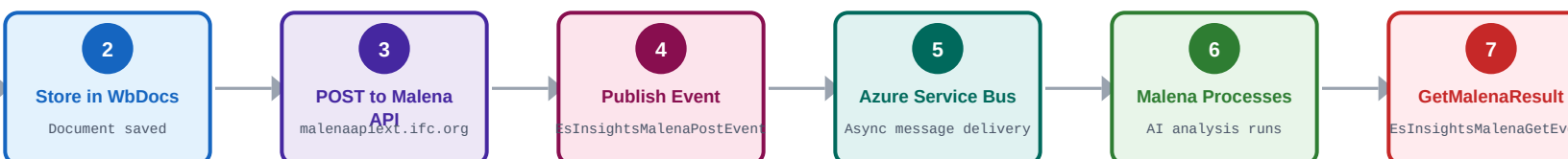
IncidentId	int	required
EsId	int	required
UploadFile	binary	document file

RETURNS: IncidentDocumentUploadDocument { documentId, wbDocsKey, STATUS_CODE: 0 }

12. ES Insights API — /api/esrd/EsInsights

The ES Insights controller integrates with **Malena AI** (IFC's machine learning service) for automated E&S; document analysis, and with **Power BI Embedded** for the analytics dashboard. Malena AI provides sentence-level risk classification against IFC Performance Standards.

Malena AI Document Analysis — Async Event Flow



POST StartMalenaUpload

[Authorize](#)

Initiate Malena AI document analysis. Fetches the document from WbDocs, sends it to the Malena AI API (malenaapiext.ifc.org) via HTTP POST with subscription key. Fires EsInsightsMalenaPostEvent domain event on Azure Service Bus for async processing. Returns upload job status.

RETURNS: SaveResponse { STATUS_CODE: 0, jobId for tracking }

POST GetMalenaResult

[Authorize](#)

Retrieve results of Malena AI document analysis. Polls for completion of the analysis job. Fires EsInsightsMalenaGetEvent. Returns analysis completion status and triggers storage of sentence-wise data.

RETURNS: SaveResponse { STATUS_CODE: 0, analysisStatus }

GET GetEsInsightDocuments

[None \(open\)](#)

List all E&S documents available for Malena AI analysis for a project. Returns document metadata for documents uploaded to the project that can be submitted for AI risk assessment.

REQUEST PARAMETERS

EsId int (query string) required
FolderKey string (query string) WbDocs folder key

RETURNS: EsInsightsDocumentsResponse { Result: List<EsInsightsDocumentInfo> }

GET

GetEsInsightsSentensewiseData

InsightsDashboardView

Retrieve sentence-level Malena AI analysis results. Returns each sentence from the analyzed document classified by: E&S risk category, IFC Performance Standard reference, risk level (High/Medium/Low), and confidence score.

REQUEST PARAMETERS

DocumentId `string` (query ~~required~~)— document to get analysis for

RETURNS: `EsInsightsSentenceWiseDataResponse { sentences[], eachSentence: { text, category, performa`

GET

GetPowerBiReportDetails

InsightsDashboardView

Get Power BI report configuration for the ES Insights embedded analytics dashboard. Returns workspace ID, report ID, and a fresh embed URL with access token.

RETURNS: `GetPowerBiReportResponse { WorkspaceId, ReportId, EmbedUrl, AccessToken, TokenExpiry }`

GET

GetPowerBiAccessToken

Authorize

Get a Power BI access token for report embedding without the full report configuration. Lightweight endpoint for token refresh.

RETURNS: `PowerBiAccessTokenResponse { AccessToken, TokenExpiry }`

13. Other API Controllers

13.1 OtherInitiatives API — /api/esrd/OtherInitiatives

Tracks WBG/IFC Cross-Collaboration initiatives and CAO Policy Action Items alongside the main project portfolio.

Method	Endpoint	Description
GET	GetOtherInitiatives	Fetch all initiatives. Optional filters via OtherInitiativesQuery. Returns List<OtherInitiativesInformation>
POST	SaveOtherInitiatives	Create or update. Body: OtherInitiativesId (0=new), Description, InitiativeTypeId, Deadline, FiscalYear
POST	AddOtherInitiativesType	Add new initiative type category for classification purposes.
POST	DeleteOtherInitiatives	Delete initiative. Body: {OtherInitiativesId: int}. Soft-delete retained in audit.
POST	DownloadOtherInitiativesReport	Download Other Initiatives Excel report. Returns FileStreamResult.

13.2 Document API — /api/esrd/Document

Manages project document records linked to the World Bank WbDocs document management system.

Method	Endpoint	Description
GET	GetDocuments	Fetch all documents for a project. Query: EsId. Returns document list with type, name, status, notes
POST	SaveDocument	Save new document record. Body: EsId, DocumentName, DocumentType, DocumentKey (WbDocs)
POST	UpdateDocumentNotes	Update notes on existing document without re-uploading. Validated by UpdateDocumentNotesCom
POST	DeleteDocument	Delete document record. Removes metadata and WbDocs link per World Bank document retention

13.3 AuditSummary API — /api/esrd/AuditSummary

Provides a complete, read-only audit trail for compliance and accountability purposes.

Method	Endpoint	Description
GET	GetAuditSummary	Fetch chronological change log for a project. Query: GetAuditSummaryQuery (EsId). Returns every

13.4 DropDownValues & MasterData APIs

Controller	Method	Endpoint	Description
DropDownValues	GET	GetDropDownValues	All UI dropdown lists in one call: Regions, Countries, Sectors, ManagementGrou

Controller	Method	Endpoint	Description
EsrdMasterData	GET	MasterDataInfo	GBV country risk tables, EHS sector guidelines, PS reference data for question
EsrdMasterData	GET	GetGbvTableData	GBV risk matrix data for the GBV scoring display matrix.
EsrdMasterData	POST	UploadGbvTableData	[FromForm] Upload updated GBV risk matrix from Excel. Replaces existing mat
EsrdMasterData	POST	DownloadGbvTableDataReport	Download current GBV risk matrix as Excel report.

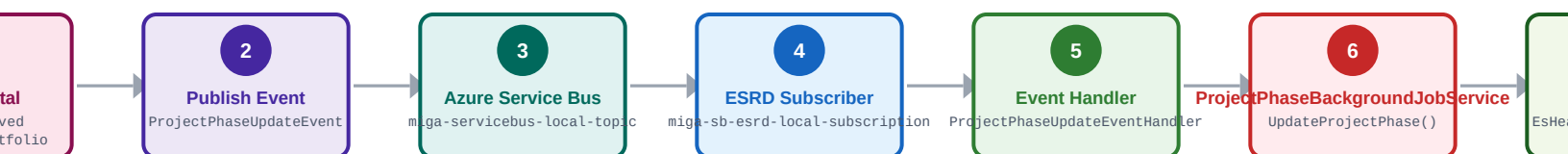
13.5 ProvideClearance & UnassignedRequests APIs

Controller	Method	Endpoint	Description
ProvideClearance	GET	ProvideClearanceProjects	List pipeline projects awaiting E&S clearance memo approval by a m
ProvideClearance	POST	ApproveProvideClearanceProject	Manager approves E&S clearance. Advances DD stage. Validated b
ProvideClearance	POST	DownloadProvideClearanceProjectsReport	Download clearance projects report as Excel.
UnassignedRequests	GET	GetUnassignedRequests	List all pending E&S specialist assignment requests.
UnassignedRequests	POST	SaveUnassignedRequests	Process/save an assignment request. Saves via SaveUnassignedR

14. Domain Events & Azure Service Bus

The MIGA E&S; Portal communicates with other MIGA microservices via **Azure Service Bus** domain events. This enables loose coupling — services don't call each other directly, they publish and subscribe to messages.

Domain Event Flow — ProjectPhaseUpdate Example



Event	Direction	Handler	Action
ProjectPhaseUpdateEvent	RECEIVED from MIGA Portal	ProjectPhaseUpdateEventHandler	Calls ProjectPhaseBackgroundJobService.UpdateProjectPhase()
EsInsightsMalenaPostEvent	RECEIVED (async result)	EsInsightsMalenaPostEventHandler	Processes Malena AI POST result. Stores analysis job results.
EsInsightMalenaGetEvent	RECEIVED (async result)	EsInsightsMalenaGetEventHandler	Processes Malena AI GET result. Stores sentence-wise insights.
TeamAssignmentApprovedEvent	PUBLISHED to Service Bus (Consumed by MIGA Portal)		Fired when team assignment is approved. MIGA Portal consumes it.
ClimateTeamInfoUpdatedInClimateEvent	RECEIVED (disabled)	Commented out	Future: sync climate team info from Climate Portal. Commented out.

■ **INFO:** All event handlers include try/catch. Failures are reported to `IDomainEventStatusNotification` which logs the failure (event name, input params, exception, timestamp) to both the Azure Service Bus failed events queue AND Azure Application Insights for monitoring.

15. FluentValidation — All 25 Validators

FluentValidation runs automatically BEFORE any handler executes, via MediatR pipeline behavior. If validation fails, HTTP 400 Bad Request is returned with error details — the handler never runs.

Validator	Command/Query Validated	Key Rules
SaveMonitoringActivityCommandValidator	SaveMonitoringActivityCommand	12 conditional date rules: if AmrReviewFlag=true → AmrFromDate
SaveMonitoringActivityAndAmrHistoryCommandValidator	SaveMonitoringActivityAndAmrHistoryCommand	EsId > 0 required. FiscalYear required. AmrFromDate + AmrToDate
SaveDueDiligenceCommandValidator	SaveDueDiligenceCommand	Nested validator: for each QuestionDetail where QuestionSection
SaveEsapActionCommandValidator	SaveEsapActionCommand	EsId > 0 required. Action description text required. DueDate requ
SaveGbvQuestionnairesCommandValidator	SaveGbvQuestionnairesCommand	EsId required. GbvId required. Questionnaires list not empty. Eac
SaveGbvActionPlanDetailCommandValidator	SaveGbvActionPlanDetailCommand	GbvId required. EsId required. TypeOfAction required.
SaveTeamInformationCommandValidator	SaveTeamInformationCommand	EsId required. ProjectNumber required. EsSpecialistUpI > 0 (leac
SaveIncidentCommandValidator	SaveIncidentCommand	EsId required. IncidentDate required. Description not empty. Inci
SaveNatureFinanceCommandValidator	SaveNatureFinanceCommand	EsId required. IsNatureFinance value required. EsStage required
SaveOtherInitiativesCommandValidator	SaveOtherInitiativesCommand	Description required. InitiativeTypeId > 0 required. FiscalYear rec
AddOtherInitiativesTypeCommandValidator	AddOtherInitiativesTypeCommand	TypeName required. TypeCode required and unique.
SaveAdditionalInformationForPipelineCommandValidator	SaveAdditionalInformationForPipelineCommand	EsId > 0 required.
SaveAdditionalInformationForPortfolioCommandValidator	SaveAdditionalInformationForPortfolioCommand	EsId > 0 required.
ApproveProvideClearanceProjectCommandValidator	ApproveProvideClearanceProjectCommand	EsId required. IsApproved boolean must be set.
DeleteEsapActionCommandValidator	DeleteEsapActionCommand	esapActionIdList must not be null or empty. At least one ID requir
DeleteGbvActionPlanDetailCommandValidator	DeleteGbvActionPlanDetailCommand	GbvActionPlanDetailId > 0 required. GbvId required.
GetEsapDetailsQueryValidator	GetEsapDetailsQuery	EsId > 0 required.
GetNatureFinanceQueryValidator	GetNatureFinanceQuery	EsId > 0 required.
GetManagersProjectsQueryValidator	GetManagersProjectsQuery	ManagerUpI > 0 required.
GetSpecialistsProjectsQueryValidator	GetSpecialistsProjectsQuery	UpI > 0 required.
UpdateDocumentNotesCommandValidator	UpdateDocumentNotesCommand	DocumentId > 0 required. Notes text not empty.
DownloadGbvTableDataReportQueryValidator	DownloadGbvTableDataReportQuery	At least one filter parameter required.
UploadGbvActionPlanDetailsCommandValidator	UploadGbvActionPlanDetailsCommand	EsId required. UploadFile not null.
UploadGbvTableDataCommandValidator	UploadGbvTableDataCommand	UploadFile not null. File extension must be .xlsx.
SaveHighRiskDetailsCommandValidator (import)	SaveHighRiskDetailsCommand	EsId required.

16. Configuration & Environment Variables

All sensitive configuration is stored in **Azure Key Vault**. Non-sensitive configuration uses environment variables defined in `launchSettings.json` (development) or set at deployment time (production). Nothing sensitive is ever stored in `appsettings.json` or code files.

16.1 Environment Variables (launchSettings.json)

Variable	Example Value	Used For
KeyVaultName	migaportal-dev-kvd	Azure Key Vault name for secrets
KvTenantId	31a2fec0-266b-4c67-b56e-2796d8f59c36	Azure AD tenant ID for Key Vault auth
WebApiAzureAdAudience	https://migaportalclouddevapi.worldbank.org	Expected JWT token audience
AzureAdInstance	https://login.microsoftonline.com/	Azure AD login endpoint
MPMIntegrationServiceUrl	https://migaportalclouddev.api.worldbank.org/	MIGA Portal Integration microservice
NotificationServiceUrl	https://migaportalclouddev.api.worldbank.org/	Email Notification microservice
AuthorizationServiceUrl	https://migaportalclouddev.worldbank.org	Authorization Service
MasterdataServiceUrl	https://migaportalclouddev.worldbank.org/	Master Data Service
OdsServiceUrl	https://miga1.esbt.worldbank.org	Operational Data Store
MalenaServiceUrl	https://malenaapiext.ifc.org/	Malena AI Service (IFC)
ServiceBusTopicName	miga-servicebus-local-topic	Azure Service Bus topic
ServiceBusSubscriber	miga-sb-esrd-local-subscription	ESRD subscription for short events
ServiceBusSubscriberForLRP	miga-sb-esrd-local-subscription-lrp	ESRD subscription for long-running events
WbdocsServiceUrl	https://wbdocsapi.ocappsdev.worldbank.org/V2/services/wbdocs	WbDocs V2 service
WbDocsI4Key	0b0f424682ae20e3	WbDocs integration key
WbDocsSiteKey	QA-MIGAGP	WbDocs site key
EsIncidentDocsFolderKey	beae758a-959d-42c3-abae-2608424f6bb6	Azure Blob folder for incident docs
FromAddress	jgopal@worldbankgroup.org	Email 'from' address for notifications
GbvSpecialistMailId	avenkatraman1@worldbankgroup.org	GBV Specialist notification email
Environment	Dev / QA / Production	Controls Swagger visibility (hidden in Production)
CacheDurationMinutes	30	Master data in-memory cache duration
PowerBiResourceUrl	https://analysis.windows.net/powerbi/api/	Power BI API resource URL
WorkspaceId	5736cac6-ecf2-4de3-8046-14aefab04634	Power BI workspace GUID

Variable	Example Value	Used For
ReportId	08b78894-fec4-455f-b98c-64f63a6945fb	Power BI report GUID
BlobStorageUrl	https://migaportaldevstrg.blob.core.windows.net/	Azure Blob Storage base URL
TableStorageUrl	https://migaportaldevstrg.table.core.windows.net/	Azure Table Storage base URL

16.2 Azure Key Vault Secrets

Secret Name	Type	Used For
DbConnStrESRD	Connection String	Full PostgreSQL connection string for the ESRD database. Fetched at startup.
app-insights-connectionstring	Connection String	Azure Application Insights connection string for production monitoring.
MalenaSubscriptionKey	API Key	Subscription key header for Malena AI service HTTP calls (Ocp-Apim-Subscription-Key).

■ **NOTE:** Key Vault secrets are fetched *ONCE* at application startup via `Secrets().GetSecretAsync()`. If Key Vault is unavailable, the application will **FAIL TO START**. Ensure the deployment environment has Key Vault access configured.

17. AutoMapper Profiles — 30 Mapping Profiles

AutoMapper profiles define bidirectional mappings between database entities, domain models, and API request/response DTOs. All 30 profiles are auto-discovered at startup via `services.AddAutoMapper(typeof(Startup))`. Each profile uses `CreateMap()` with explicit `.ForMember()` overrides.

Profile File	Primary Mappings	Key Notes
MonitoringMapperProfile	SaveMonitoringActivityCommand → EsMonitoring; EsMonitoring → SaveMonitoringActivityCommand	Date fields converted to DateTimeOffset; EsMonitoring to SaveMonitoringActivityCommand
DueDiligenceMapperProfile	EsHeader → DueDiligenceQuestionnaire; PsQuestionMasterCompResBaseQuestionnaire; PsMappingQuestionnaire	Complex base questionnaire mapping; PsMappingQuestionnaire
EsapActionMapperProfile	SaveEsapActionCommand → EsapAction; EsapAction → SaveEsapActionCommand	EsapAction mapping includes EsapActionHistory; EsapActionHistory
PortfolioProjectsTabMapperProfile	PortfolioProjectsEntityModel → PortfolioProjectInformation	Main portfolio mapping — 80+ field mapping including
EsHeaderPipelineProjectMapperProfile	EsHeader → PipelineProjectInformation	Pipeline project fields including DD stage, team info
DashboardMapperProfile	Various entity models → DashboardDetailsResponse object	Aggregation and calculation for chart data.
ReportsMapperProfile	Report entity models → Report response DTOs	Used by download report services for data preparation.
GbvDetailMapperProfile	GBV questionnaire entities → GbvDetailsResponse	GBV risk scoring calculation during mapping.
GbvActionPlanDetailMapperProfile	SaveGbvActionPlanDetailCommand → GbvActionPlanDetail entity	Action plan CRUD mapping.
HighRiskMapperProfile	HighRiskQuestions → HighRiskBaseQuestionnaire; HighRiskBaseQuestionnaire → HighRiskQuestions	3 Mapping Questions shield highRiskMappingBaseQue
IncidentMapperProfile	SaveIncidentCommand → Incident entity; Incident → SaveIncidentCommand	IncidentDetailsResponse detail sub-records mapping.
NatureFinanceMapperProfile	SaveNatureFinanceCommand → NatureFinance entity; NatureFinance entity → SaveNatureFinanceCommand	Upper Finance storage reference detailed.
OtherInitiativesMapperProfile	SaveOtherInitiativesCommand → OtherInitiatives entity	Basic CRUD mapping.
TeamInformationMapperProfile	SaveTeamInformationCommand → EsTeamInformation entity	Specialist UPI list handling.
AmrHistoryMapperProfile	AMR History entities ↔ AmrHistoryDetailsResponse	Date formatting.
AuditSummaryMapperProfile	Audit log entities → AuditSummaryResponse	Timestamp formatting.
DocumentMapperProfile	SaveDocumentDetailsModel → Document entity	Document metadata mapping.
ESInsightMapperProfile	ES Insight entities → ESInsightsDocumentsResponse	Malena AI document list mapping.
ProjectDetailsMapperProfile	ProjectDetails responses from Integration → ProjectDetailsResponse	Integration service data mapping.
ProvideClearanceProjectsMapperProfile	ProvideClearance entities → ProvideClearanceProjectsResponse	Clearance workflow mapping.
ShelvedProjectsMapperProfile	EsHeader → ShelvedProjectInformation	Shelved status code mapping.
UnassignedProjectMapperProfile	Unassigned entities → UnassignedRequestsResponse	Assignment request mapping.
DropDownMapperProfile	Master data entities → DropDownValuesResponse	Lookup data flattening.
PipelineDetailsToPipelineProjectMapperProfile	Integration pipeline data → PipelineProjectInformation	Cross-source merge mapping for pipeline.
PipelineDetailsToPortfolioProjectMapperProfile	Integration portfolio data → PortfolioProjectInformation	Cross-source merge mapping for portfolio.

Profile File	Primary Mappings	Key Notes
ProjectAndApplicationDetailsResponseMapperProfile	Project details response → Project details model	Application data mapping.
GetGbvTableDataMapperProfile	GBV table data entities → GbvTableDataResponse	Risk matrix mapping.
EsrdMapperProfile	General ESRD entity mappings	Shared mappings used across multiple services.
SaveAdditionalInformationForPipelineEsHeadMapperProfile	SaveAdditionalInformationForPipelineCommand → EsHead	Additional field mapping.
SaveAdditionalInformationForPortfolioEsHeadMapperProfile	SaveAdditionalInformationForPortfolioCommand → EsHead	Portfolio specific additional fields.

18. Glossary of Technical Terms

Term	Definition
AMR	Annual Monitoring Report — submitted by guarantee holders annually documenting E&S performance against ESAP
Aspose.Cells	Commercial library used for Excel (.xlsx) report generation on the server
Autofac	Advanced DI container — overrides default ASP.NET Core DI, provides more features
AutoMapper	Library that automatically copies properties between two object types by name matching
CAO	Compliance Advisor Ombudsman — independent IFC/MIGA oversight handling E&S complaints
CQRS	Command Query Responsibility Segregation — architectural pattern separating reads from writes
DD	Due Diligence — the E&S assessment conducted on pipeline projects before guarantee issuance
DI	Dependency Injection — framework automatically creates and injects class dependencies
ESAP	Environmental & Social Action Plan — contractual actions guarantee holders must complete
ESDD	Environmental & Social Due Diligence — the full assessment process for pipeline projects
ESMS	Environmental and Social Management System — risk framework at project/enterprise level
ESRR(D)	Environmental & Social Risk Rating at Due Diligence stage — rates pipeline projects
ESRR(M)	Environmental & Social Risk Rating at Monitoring stage — rates portfolio projects
EsStage	Project stage code: 'DD' (Due Diligence) or 'MN' (Monitoring)
EsId	Internal E&S Portal integer ID for each project (primary key in es_header table)
FluentValidation	Library for declaring input validation rules in C# using a fluent syntax
GBV	Gender-Based Violence — risk category assessed for all MIGA-guaranteed projects
Handler	MediatR handler class that receives a Query or Command and delegates to one Service method
IFC Track B	IFC categorization for Financial Intermediary projects requiring enhanced E&S monitoring
Key Vault	Azure Key Vault — secure secret store for database strings, API keys, subscription keys
Malena AI	IFC's AI document analysis service providing sentence-level E&S risk classification
MediatR	Open-source .NET library implementing the Mediator pattern for CQRS message dispatch
MIGA	Multilateral Investment Guarantee Agency — World Bank Group member providing PRI
MSV	Monitoring Site Visit — physical or virtual visit to a project site for E&S verification
Polly	Resilience library providing HTTP retry policies — used for Malena AI 3-retry
PolicyAuthorize	Custom World Bank attribute combining Azure AD JWT auth + ESRD permission check
PS1–PS8	IFC Performance Standards 1–8 — E&S risk management standards for MIGA projects

Term	Definition
Service Bus	Azure Service Bus — messaging backbone for domain events between MIGA microservices
UPI	Unique Person Identifier — World Bank's internal user ID integer
WbDocs	World Bank document management system for storing confidential project documents

19. Developer Guide — Adding New Features

Every new feature (endpoint) must follow the same pattern. The CORRECT order matters because each layer depends on the previous one. Follow these steps exactly:

Step	File to Create	Location	What to Do
1. Define NewFeatureResponse	NewFeatureResponse.cs	Responses/Domain/	Create class inheriting ResponseBase. Add Result field with the data type you want to return.
2. Create NewFeatureQuery or NewFeatureCommand	NewFeatureQuery.cs or NewFeatureCommand.cs	Queries/Domain/ or Commands/Domain/	Create class implementing IRequest<NewFeatureResponse>. Add all input properties and validation logic.
3. Write NewFeatureCommandValidator	NewFeatureCommandValidator.cs	Validators/	Create AbstractValidator<NewFeatureCommand> with RuleFor() calls. Use .When() for conditional rules.
4. Define NewFeatureService	NewFeatureService.cs	Interfaces/	Define the method signature: Task<ReturnType> MethodName(params). This is the contract for the service.
5. Implement NewFeatureService	NewFeatureService.cs	Services/	Implement INewFeatureService. Constructor-inject repositories and other services. V
6. Register Startup.cs	Startup.cs	ConfigureServices()	Add: services.AddTransient<INewFeatureService, NewFeatureService>(); MUST be
7. Create NewFeatureQueryHandler or NewFeatureCommandHandler	NewFeatureQueryHandler.cs or NewFeatureCommandHandler.cs	Handlers/Domain/	Implement IRequestHandler<TQuery, TResponse>. Constructor-inject INewFeatureS
8. Add MapFeatureProfile	MapFeatureProfile.cs	Mapping/	Add CreateMap<Source, Destination>() with ForMember() overrides. MediatR auto-c
9. Add [EndpointName]Controller	[EndpointName]Controller.cs	Controllers/	Add [HttpGet/Post] method with [PolicyAuthorize]. Call _mediator.Send(query/command)
10. Test API Swagger locally	API Swagger	/api/esrd/swagger	Open Swagger UI. Authenticate with Bearer token. Find new endpoint. Test with sam

✓ **PATTERN:** The handler should *ONLY* call the service and wrap the response. *ALL* business logic, *ALL* validation logic, *ALL* data transformation goes in the Service. Controllers dispatch to MediatR and nothing else.

■ **NOTE:** Never inject a Service directly into a Controller. Never inject a Repository directly into a Handler. The Controller → Handler → Service → Repository chain must be respected.